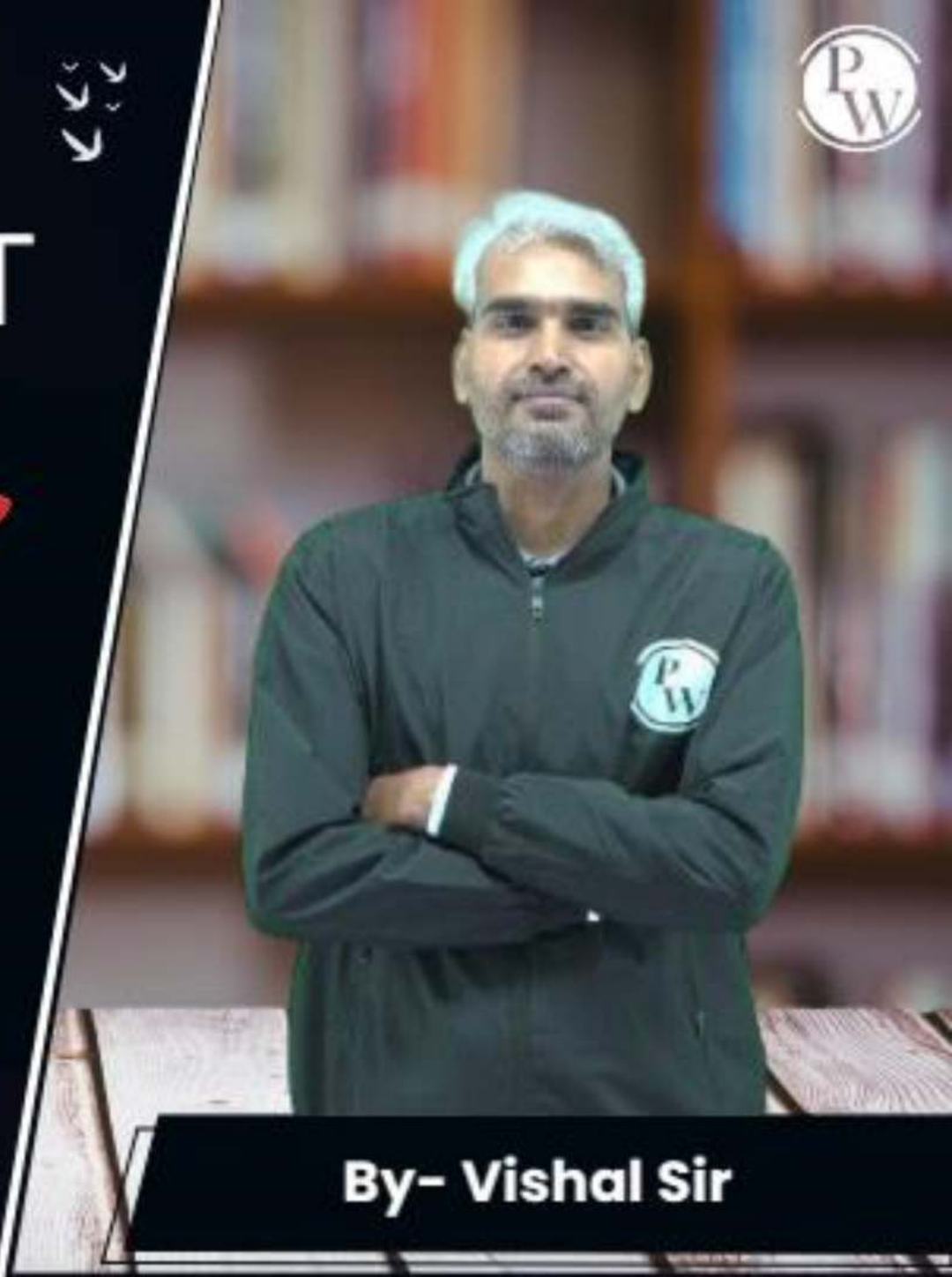
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 24



By- Vishal Sir

Recap of Previous Lecture



Topic — Order of an element in a group

Topic — Sub-group

Topic — Cyclic group

Slide

Topics to be Covered



Topic Cyclic group

Topic Some important properties

Slide



Topic : Cyclic group

* Let $(G, *)$ is a group. If there exists any element $a \in G$ such that Every element of set G can be represented in the form $(a)^n$ for some positive integer 'n', then

$(a)^n$ = $\underbrace{a * a * \ldots * a}_{n \text{ time } a}$

'n' : Can not be '0'.

group $(G, *)$ is called a Cyclic group. and element 'a' is one of the generator of cyclic group $(G, *)$

Note: In a cyclic group, Number of generators may be more than '1' as well

Note: Let $(G, *)$ be the group and 'e' is the identity element of the group.

$(e)^1 = e$

$(e)^2 = e * e = e$

$(e)^3 = e^2 * e = e * e = e$

⋮

Identity element can not generate any other element except itself.

∴ Identity element can never be a generator of the set containing any other element apart from itself

Q: $\{1, -1\}$ is a group w.r.t. multiplication

Order of group = 2

Here 1 = identity $\therefore$ can not generate any other element except itself

- $(-1)^1 = -1$
- $(-1)^2 = 1 = e$

} all the element of the group can be generated by '-1'

$\Downarrow$

Order $(-1) = 2$

$\therefore$ $\{1, -1\}$ w.r.t. multiplication is a cyclic group and '-1' is the generator of the cyclic group.

① Check whether the group $\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group or not, if cyclic then find all the generators of the cyclic group

$O(G) = 3$

1 = identity (Not a generator)

$O(1) = 1 \neq O(G)$

$(\omega)^1 = \omega$

$(\omega)^2 = \omega^2$

$(\omega)^3 = \omega^3 = 1$ = identity

$O(\omega) = 3 = O(G)$

All elements of set

$\therefore \omega$ is a generator

$(\omega^2)^1 = \omega^2$

$(\omega^2)^2 = \omega$

$(\omega^2)^3 = 1$ = identity

$O(\omega^2) = 3 = O(G)$

All elements of set

$\therefore \omega^2$ is a generator

$\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group, and ω & ω^2 are the generators

(2) Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$O(G) = 4$

1 :- identity (Not a generator)

$O(1) = 1 \neq O(G)$

$(-1)^1 = -1$

$(-1)^2 = 1$ } = identity

$O(-1) = 2 \neq O(G)$ →

$(-1)^3 = -1$ } Repeat

$(-1)^4 = 1$ } = identity

$(-1)^5 = -1$ } Repeat

$(-1)^6 = 1$ } = identity

If order of an element is not Equal to the order of the group, then it will generate the identity element before it has generated all the elements of the set. And after that if we increas the Power then it will start repeating the elements that it has already generated and will not be able to generate the remaining elements so such an element can never be the generator.

(2) Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$O(G) = 4$

1 :- identity (Not a generator)

$O(1) = 1$

$(-1)^1 = -1$

$(-1)^2 = 1$ = identity

$(-1)^3 = -1$

$(-1)^4 = 1$

$(-1)^5 = -1$

$(-1)^6 = 1$

elements will be repeated

$O(-1) = 2 \neq O(G)$

$(i)^1 = i$

$(i)^2 = i^2 = -1$

$(i)^3 = i^2 \cdot i = -i$

$(i)^4 = i^2 \cdot i^2 = 1$ = identity

$O(i) = 4 = O(G)$

All elements

$\therefore$ 'i' is a generator

$(-i)^1 = -i$

$(-i)^2 = i^2 = -1$

$(-i)^3 = +i$

$(-i)^4 = 1$ = identity

$O(-i) = 4 = O(G)$

All elements

$\therefore$ '$-i$' is also a generator

$\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group & 'i' & '$-i$' are generators

H.W. ③ Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not. If cyclic then find all the generators of the cyclic group $(O(G)=4)$

eg: ③ Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not, if cyclic then find all the generators of the cyclic group $(O(G)=4)$

1 = identity (Not a generator)

$O(1) = 1 \neq O(G)$

∴ Can not be the generator

$(3)^1 = 3$

$(3)^2 = 3 \otimes_8 3 = 1$ identity

$O(3) = 2 \neq O(G)$

∴ Can not be the generator

$(5)^1 = 5$

$(5)^2 = 5 \otimes_8 5 = 1$ (Identity)

$O(5) = 2 \neq O(G)$

∴ Can not be the generator

$(7)^1 = 7$

$(7)^2 = 7 \otimes_8 7 = 1$ (identity)

$O(7) = 2 \neq O(G)$

∴ Can not be the generator

No generator, ∴ Group $\{1, 3, 5, 7\}$ w.r.t $\otimes_8$ is not a cyclic group



Topic : Properties w.r.t. Cyclic group

① Identity element can not be the generator of the group containing any other element except itself



Topic : Properties w.r.t. Cyclic group

② Let $(G, *)$ be a group of order $= |G|$

if there exist some element $a \in G$

such that $O(a) = |G| =$ order of group

then $(G, *)$ is a cyclic group

and 'a' is one of its generator



Topic : Properties w.r.t. Cyclic group

③ If $a \in G$ is the generator of cyclic group $(G, *)$ then "$inv(a) = a^{-1}$" will also be the generator of that cyclic group

We know $O(a^{-1}) = O(a)$

$\therefore O(a^{-1}) = O(G)$

Hence a^{-1} is also the generator



Topic : Properties w.r.t. Cyclic group

④ Let $(G, *)$ be a group of order $= |G|$

if there exist no element $a \in G$

s.t $O(a) = |G| =$ order of group

then group is not a cyclic group

ie, if $O(a) \neq O(G)$, $\forall a \in$ group $(G, *)$

then $(G, *)$ is not a cyclic group

Example on next Slide.

eg: Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not, if cyclic then find all the generators of the cyclic group $(O(G)=4)$

1 = identity (Not a generator)

$O(1) = 1 \neq O(G)$

∴ Can not be the generator

$(3)^1 = 3$

$(3)^2 = 3 \otimes_8 3 = 1$ identity

$O(3) = 2 \neq O(G)$

∴ Can not be the generator

$(5)^1 = 5$

$(5)^2 = 5 \otimes_8 5 = 1$ (Identity)

$O(5) = 2 \neq O(G)$

∴ Can not be the generator

$(7)^1 = 7$

$(7)^2 = 7 \otimes_8 7 = 1$ (identity)

$O(7) = 2 \neq O(G)$

∴ Can not be the generator

$\forall a \in \{1, 3, 5, 7\}$ $O(a) \neq O(G)$ w.r.t $\otimes_8$

∴ Not a cyclic group.

Q:- Let $\{0, 1, 2, 3, 4\}$ w.r.t. binary opⁿ $\oplus_5$ is a cyclic group, then find all generators of the cyclic group

Order of group = 5 (Prime no.)

∴ for Every non-identity element

$O(\text{element}) = O(G)$

0 = identity

∴ '0' can not be the generator

$O(\text{identity}) = 1$

∴ $O(1) = O(2) = O(3) = O(4) = O(G) = 5$

∴ 1, 2, 3 & 4 are generators of group.

$\{1, 2, 3, 4, 5, 6\}$ is a group w.r.t. $\otimes_7$

Check whether the group is a cyclic group or not. If cyclic then find all generators of the cyclic group

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary op^n $\otimes_7$ be a Cyclic group. Find all the generator of Cyclic group.

Order of group = 6

1 = identity
∴ 1 is not a generator

$(3)^1 = 3$
$(3)^2 = 2$
$(3)^3 = 6$
$(3)^4 = 4$
$(3)^5 = 5$
$(3)^6 = 1$ = identity

all elements

∴ '3' is a generator

inv(3) = 5
∴ '5' is also the generator

$(2)^1 = 2$
$(2)^2 = 4$
$(2)^3 = 1$ = identity

$O(2) = 3 \neq O(G)$
∴ Not a generator

inv(2) = 4, ∴ $O(4) = O(2) = 3 \neq O(G)$
Hence '4' is not a generator

$(6)^1 = 6$
$(6)^2 = 1$ = identity
$O(6) = 2 \neq O(G)$
∴ 6 is not a generator



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G,*)$ be a cyclic group of order $= n$, and $a \in G$ is one of its generator then

$$\{ (a)^m \mid m \in \text{Natural no. } (N = \{1,2,\dots\}) \text{ and } m < n \text{ and } GCD(m,n) = 1 \}$$

this set is the set of all generator of Cyclic group G.

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary op^n $\otimes_7$ be a Cyclic group with element '3' as one of its generator. Find all generators of the cyclic group

$\Rightarrow$ order of group $= 6 \Rightarrow$ Positive integers that are less than '6' are Co-prime to '6' are '1' & '5'

* If '3' is a generator then

$\{(3)^1, (3)^5\}$ is the set of all generators of the cyclic group

$\{3, 5\}$ is the set of all generators.



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G,*)$ be a cyclic group of order $= n$, and $a \in G$ is one of its generator then

$$\{(a)^m \mid m \in \text{Natural number} \text{ and } m < n \text{ (order of group)} \text{ and } GCD(m,n) = 1\}$$

is the set of all the generator of the cyclic group

⑥ Let $(G,*)$ be a cyclic group of order $= n$, then number of generators of the cyclic group will be same as number of positive integers that are less than 'n' and Co-prime to 'n'

Q: Let $(G, *)$ be a cyclic group of order = 50, find the number of generators of cyclic group $(G, *)$

- Number of generators of the cyclic group will be same as no. of +ve integers that are less than 50 & Coprime to 50.
- Such numbers are, 1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39, 41, 43, 47, 49

'20' such Positive integers are there.

∴ No. of generators = 20

✡ Euler's totient function $\phi(n)$

- Let 'n' is any fixed positive integer, then number of positive integers that are less than 'n' and Coprime to 'n' can be given by Euler's totient function $\phi(n)$

$$\phi(n) = n \cdot \left\{ \left(1 - \frac{1}{P_1}\right) \cdot \left(1 - \frac{1}{P_2}\right) \cdot \left(1 - \frac{1}{P_3}\right) \cdots \right\}$$

$P_1, P_2, P_3, \ldots$ etc are distinct prime factors of 'n'.

Q: Let $(G, *)$ be a cyclic group of order = 50, find the number of generators of cyclic group $(G, *)$

Order of group = 50

∴ Number of generators = No. of positive integers that are less than 50 & Coprime to 50

∴ Number of generators = $\phi(50)$

$50 = 2 * 25$

$= (2)^1 * (5)^2$

2 & 5 are the only two distinct prime factors of 50!

∴ $\phi(50) = 50 * \left\{ \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right) \right\}$

$= 50 * \frac{1}{2} * \frac{4}{5} = 20$

Ans

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary oph $\otimes_7$ be a Cyclic group. Find the number of generators of above cyclic group

order $= 6$

$\therefore$ No. of generators $= \phi(6) = 6 \times \left\{\left(1-\frac{1}{2}\right)\cdot\left(1-\frac{1}{3}\right)\right\}$

$6 = (2)^1 * (3)^1$

Prim. factors

$= 6 * \frac{1}{2} * \frac{2}{3}$

$= 2$

Q: Let $(G, *)$ be a cyclic group of prime order = 'p' find the number of generators of the group

* If 'p' is a prime no., then no. of natural nos that are less than 'p' and Co-prime to p are $(p-1)$

∴ No. of generators $= (p-1)$

$$\#\text{ generators} = \phi(p) = p * \left\{1 - \frac{1}{p}\right\}$$

$$= p * \left(\frac{p-1}{p}\right) = (p-1)$$

* (If 'p' is prime no. then only prime factor is 'p'.)



Topic : Some important properties

1. Every group of prime order is a Cyclic group.
2. Every Cyclic group is abelian group (Converse need not be true

Let $(G,*)$ be a group with 'a' as one of its generator let elements $P, q \in G$. and there exists two least positive integers m & n such that $(a)^m = p$ & $(a)^n = q$

Now, $P * q = (a)^m * (a)^n$

$= (\underbrace{a*a*\cdots*a}_{m \text{ times } 'a'}) * (\underbrace{a*a*\cdots*a}_{'n' \text{ times } a})$

$= \underbrace{a*a*\cdots*a*a*a*\cdots*a}_{'m+n' \text{ times } a}$

$= (\underbrace{a*a*\cdots*a}_{'n' \text{ times } 'a'}) * (\underbrace{a*a*\cdots*a}_{'m' \text{ times } a})$

$= (a)^n * (a)^m$

$= q * p$

$\Rightarrow$ '*' is Abelian

③ Every group of prime order is abelian

{By property ① & property ②}



Topic : Some important properties

④ Every subgroup of a cyclic group is also a cyclic group, but generators of the cyclic subgroup need not be same as generator of original cyclic group

Eg: $\{1, -1, i, -i\}$ is a cyclic group with i & $-i$ as generator

$\{1, -1\}$ is a sub-group of above group, it is also cyclic, but generator is '-1'

Note: If group is cyclic then its sub-group will also be cyclic.

But, if group is not cyclic, then sub-group may or may not be cyclic



Topic : Some important properties

⑤ Every group of order ≤ 3 is always cyclic



↓

in this case identity element is the only element of the set, and it can generate itself.

∴ Cyclic

Both are prime no.

∴ Cyclic



Topic : Some important properties

⑥ Every group of order = 4 is cyclic
Except for the groups of 'Klein-4' type

Properties of "Klein-4" group:

1. order of group = 4
2. Every element is inverse of itself
3. Binary op^{n} on two non-identity elements results in third non-identity element

eg: $\{e, a, b, c\}$ (e: identity; a, b, c: non-identity)

i.e. $inv(e) = e$
$inv(a) = a$
$inv(b) = b$
$inv(c) = c$

i.e. $a * b = b * a = c$
$a * c = c * a = b$
$b * c = c * b = a$

Klein-4 groups are not cyclic but they are abelian

eg: $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is Klein-4 type group:

1. order = 4
2. $inv(1) = 1$, $inv(5) = 5$
   $inv(3) = 3$, $inv(7) = 7$
3. $3 \otimes_8 5 = 5 \otimes_8 3 = 7$
   $3 \otimes_8 7 = 7 \otimes_8 3 = 5$
   $5 \otimes_8 7 = 7 \otimes_8 5 = 3$



Topic : Some important properties

(6) Every group of order = 4 is cyclic Except for the groups of 'Klein-4' type

(7) 'Klein-4' group of order = 4 is not cyclic but it is an abelian group

(8) Every group of order ≤ 5 is cyclic Except for 'Klein-4' group of order = 4

(9) Every group of order ≤ 5 is always abelian group

Q:- $Z_n = \{0, 1, 2, \ldots\ldots, n-1\}$ is a group w.r.t. binary op^n $\oplus_n$.

In the group $Z_2 \times Z_3$ how many elements are their own inverse.

$Z_2 = \{0, 1\}$ is a group w.r.t. $\oplus_2$

$Z_3 = \{0, 1, 2\}$ is a group w.r.t. $\oplus_3$

$$Z_2 = \{0, 1\}$$

$$Z_3 = \{0, 1, 2\}$$

$$Z_2 \times Z_3 = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}$$

$Z_2 \times Z_3$ is a group

We can observe identity element w.r.t group $Z_2 \times Z_3$ is $(0,0)$

$$Z_2 \times Z_3 = \{(a,b) \mid a \in Z_2 \text{ \& } b \in Z_3\}$$

- If $(p,q), (r,s) \in Z_2 \times Z_3$

  then $(p,q) *_{\uparrow} (r,s) = (p \oplus_2 r, q \oplus_3 s)$

  Binary op^n w.r.t. group $Z_2 \times Z_3$

$$(0,1) * (1,2) = (1,0)$$

$1 \oplus_3 2$

$0 \oplus_2 1$

$Z_2 = \{0, 1\}$

$Z_3 = \{0, 1, 2\}$

$Z_2 \times Z_3 = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}$

$(0,0) * (0,0) = (0,0) = \text{identity}$

$(0,1) * (0,1) = (0,2)$

$(0,2) * (0,2) = (0,1)$

$(1,0) * (1,0) = (0,0) = \text{identity}$

$(1,1) * (1,1) = (0,2)$

$(1,2) * (1,2) = (0,1)$

Two elements are inverse of each other in $Z_2 \times Z_3$

W.r.t. above question:

How many elements in group Z_2 are inverse of their own?

$Z_2 = \{0, 1\}$ Both '0' & '1' are inverse of their own w.r.t $\oplus_2$

How many elements in group Z_3 are inverse of their own?

$Z_3 = \{0, 1, 2\}$ '0' is inverse of its own.

Only '1' element is inverse of its own

then No. of elements in group $Z_2 \times Z_3$ that are inverse of their own

= No. of elements in Z_2 that are their own inverse * No. of elements in Z_3 that are their own inverse

$= 2 * 1 = 2$

Note: Let S_1 is a group such that 'm' elements are inverse of their own

&

Let S_2 is a group such that 'n' elements are inverse of their own

then in the group $S_1 \times S_2$, 'm×n' elements will be their own inverse

Note: Let S_1 is a group such that 'm' elements are inverse of their own

&

Let S_2 is a group such that 'n' elements are inverse of their own

&

Let S_3 is a group such that 'p' elements are inverse of their own

then in the group '$S_1 \times S_2 \times S_3$', '$m.n.p$' elements will be their own inverse



2 mins Summary

Topic Cyclic group

Topic Some important properties

Slide

THANK - YOU